CS4248 NATURAL LANGUAGE PROCESSING

CS4248-11

MEME REPLY SELECTION

MEMBERS: Lavanya Garg, Rohan Sapra, Darren Zhu, Low Hui Yen, Kuvawala Mustafa Abbas

PROJECT MENTOR: Ge Xiao Yu

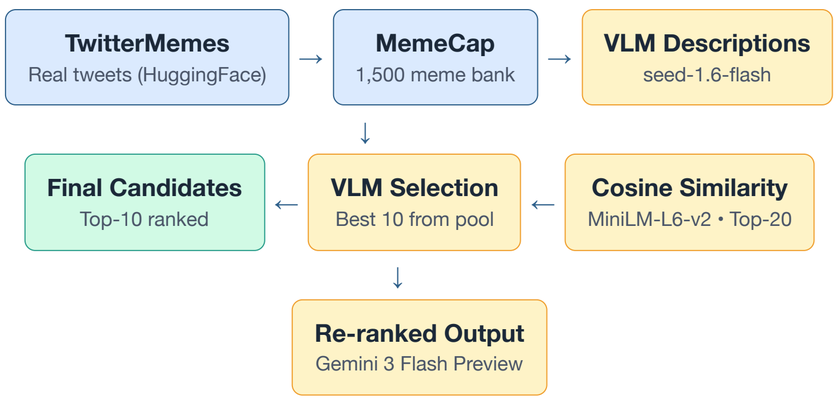
VOTE FOR US

1 ABSTRACT

Choosing the right meme reply to a post depends on both content and context. We study meme recommendation as a **ranking task**: given a tweet and candidate memes, rank the most suitable meme highest. We compare **text-only**, **image-only**, and **multimodal** approaches. The image model performs best on validation, while text is more consistent on test. The multimodal model does not outperform simpler baselines, as text benefits from rich metadata while image helps when post text is short. The modalities are thus **complementary**.

2 DATA PIPELINE

A VLM describes each meme semantically, then sentence embeddings build a candidate pool via cosine similarity.

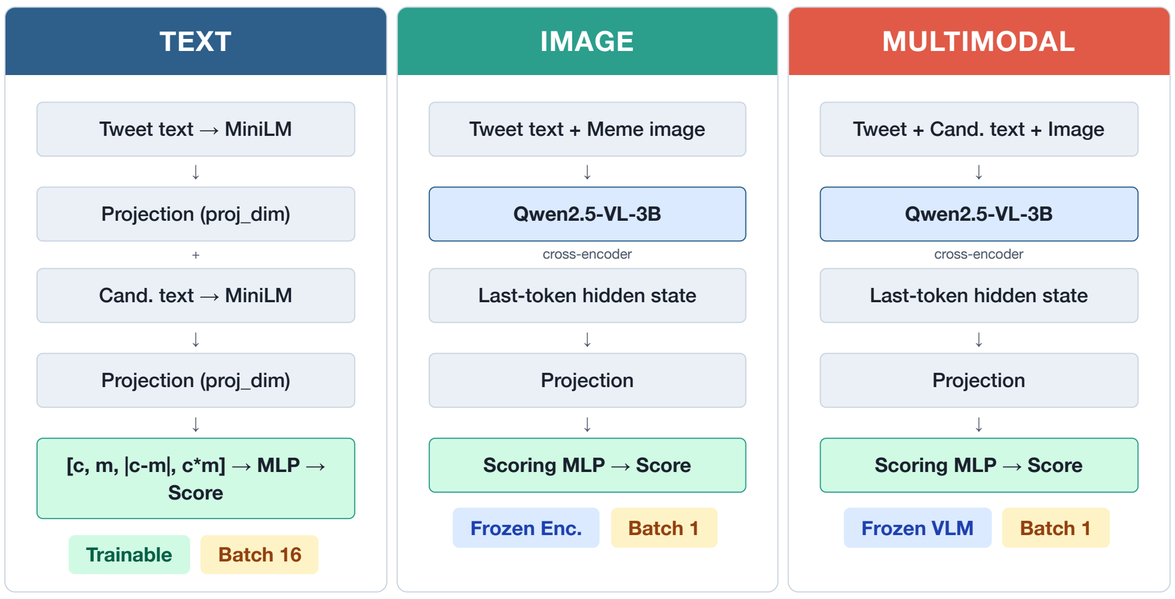


DATASET STATISTICS

96 Validation tasks	**98** Test tasks	**~9** Avg. candidates/task
~300 Avg. candidate text length (chars)	**92%** Reddit meme images (i.redd.it)	**80/10/10** Train / Val / Test split

3 MODEL ARCHITECTURE

Three ranking pipelines, all using BPR (Bayesian Personalized Ranking) loss with AdamW optimizer, 100-step warmup, and LR 2e-5.



Candidate text = meme_title + img_captions + meme_captions + metaphors. Text pipeline uses dual encoding with a Preference MLP that combines [c, m, |c-m|, c*m]. Qwen pipelines are cross-encoders: tweet (+ candidate text for multimodal) and image are processed jointly. Scoring is via projection + 2-layer MLP. Mixed-precision (BF16/FP16 via AMP auto) on CUDA.

WHY MULTIMODAL UNDERPERFORMED

- **Noisy added text:** Injected candidate text may distract the VLM from the image.
- **Frozen encoder:** Only the projection + scoring MLP is trainable, far less adaptation capacity than the text pipeline.
- **Loss near random:** Both Qwen pipelines finish at ~0.692 BPR loss (random baseline ~0.693).
- **Prompt-level fusion only:** No learned fusion module; relies entirely on the frozen VLM's internal attention.

4 RESULTS & ANALYSIS

EVALUATION METRICS

- **Recall@1:** fraction of tasks where the top-ranked candidate is the gold meme
- **MRR:** average of 1/rank of the first correct meme
- **nDCG@10:** ranking quality across the full top-10 list, weighted by position
- **Score@1:** normalized inverse gold rank of the top prediction

Split	Pipeline	Recall@1	MRR	nDCG@10	Score@1
Val	Text	0.2500	0.3891	0.7440	0.4392
	Image	0.2708	0.4276	0.7557	0.4426
	Multimodal	0.1979	0.3570	0.7195	0.3652
Test	Text	0.2143	0.3853	0.7457	0.4189
	Image	0.2143	0.3886	0.7315	0.3804
	Multimodal	0.1837	0.3193	0.7101	0.3745

PERFORMANCE BY DATA CONDITION (RECALL@1)

Condition	Text R@1	Image R@1	Multi R@1
Short tweets (val)	11.5%	25.0%	7.7%
Short tweets (test)	13.5%	18.9%	8.1%
Weak cand. text (≤260 chars, val)	5.3%	21.1%	10.5%
Rich metadata (metaphors 100%, val)	35.7%	23.8%	21.4%
Long cand. text (301+ chars, test)	34.0%	23.4%	27.7%
MeMe Live subset (val)	64.3%	50.0%	64.3%
Non-MeMe Live subset (val)	18.3%	23.2%	12.2%

5 KEY FINDINGS

1. **Text is a semantic metadata pipeline.** Titles, captions, and metaphors give a dense meme summary without image interpretation.
2. **Image outperforms on underspecified tweets.** Visual template cues capture information that text alone cannot recover.
3. **Clear complementarity.** Oracle (text+image) = 37.5% val, 27.6% test. Oracle (all 3) = 41.7% val, 31.6% test.
4. **Multimodal defaults to high-frequency candidates.** 35-45% of errors predict generic memes. Frozen VLM fusion lacks adaptation capacity.

ERROR ANALYSIS

Near-Miss Rate (rank 2-3 when wrong)

6 CONCLUSION & FUTURE WORK

KEY TAKEAWAYS

- Text and image solve different problem slices; neither modality dominates
- Text excels with rich metadata; image excels on underspecified tweets
- 67/98 test tasks unsolved by all models, confirming cross-modal complementarity

FUTURE DIRECTIONS

- **LORA / PARTIAL UNFREEZING** of Qwen for multimodal adaptation
- **TEXT FIELD ABLATION** to quantify each metadata field's contribution
- **LEARNED FUSION MODULE** outside the frozen VLM
- **CONFIDENCE-AWARE SELECTION:** defer to text when margins are low

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